BIOLOGY

74. Which one of the following parts of the pitcher plant becomes modified into a pitcher?

(a) Stem (b) Leaf
(c) Stipule (d) Petiole

Ans. (b) : In pitcher plant, the lamina of leaf is modified into pitcher and the leaf apex gives rise to a coloured lid for attracting the insects.

75. In human body, which one of the following hormones regulates blood calcium and phosphate?

(a) Glucagon
(b) Growth hormone
(c) Parathyroid hormone
(d) Thyroxin

Ans. (c) : Parathyroid hormone secrets from parathyroid gland. This hormone regulates calcium and phosphate level in blood. Parathyroid hormone raises the level of calcium in the blood and decreases the level of phosphorous in the blood.

76. How do most insects respire?

(a) Through skin
(b) Through gills
(c) By lungs
(d) By tracheal system

Ans. (d) : Most insects respire through tracheal system that lies on a network of small tubes that channel oxygen directly to the different part of the body. The tracheal system is composed of chitin-ringed tubes called trachea that connect directly to the air through opening in the body wall called spiracles.

77. In human beings, normally in which one of the following parts, does the sperm fertilize the ovum?

(a) Cervix
(b) Fallopian tube
(c) Lower part of uterus
(d) Upper part of uterus

Ans. (b) : Fallopian tube is a tube like structure that carries egg from the ovary to uterus. In this tube the sperm meets ovum and fertilization occurs.

78. Which one of the following parts of human brain is the regulating centre for swallowing and vomiting?

(a) Cerebellum
(b) Cerebrum
(c) Medulla oblongata
(d) Pons

Ans. (c) : The medulla oblongata is a major structure located in the lower half of the brainstem. It is responsible for the regulation of heart rate, breathing and blood pressure, as well as reflexes, such as vomiting, sneezing and coughing. This section of the brain helps transfer messages to the spinal cord from the body and the thalamus, which is in the brain. The main function of the thalamus is to process information to and fro the spinal cord and the cerebellum. The medulla oblongata helps regulate breathing, heart and blood vessel function, digestion, sneezing and swallowing. This part of the brain is a center for respiration and circulation. Sensory and motor neurons (nerve cells) from the forebrain and midbrain travel through the medulla.

79. Production of which one of the following is a function of the liver?

(a) Lipase (b) Urea
(c) Mucus (d) Hydrochloric acid

Ans. (b) : Urea is the main nitrogenous excretory product of Ureotelic animals, produced by liver cells from de-aminated excess amino-acids via the urea cycle.

80. Which one of the following is not a digestive enzyme in the human system?

(a) Trypsin (b) Gastrin
(c) Ptyalin (d) Pepsin

Ans. (b) : Gastrin is a hormone, which stimulates secretion of digestive enzyme in stomach. Trypsin, Pepsin and Ptylin are the digestive enzymes of human system.

81. In the human body, which structure is the appendix attached to?

(a) The large intestine
(b) The small intestine
(c) The gall bladder
(d) The stomach

Ans. (a) : The appendix is a small tube-shaped structure attached to the first part of the colon, also known as the colon.

82. Robert Webster is known for his work associated with which one of the following?

(a) Cardiology (b) Influenza virus
(c) HIV/AIDS (d) Alzheimer

Ans. (b) : Robert Gordon Webster is an avian influenza authority who correctly posited that pandemic strains of flu arise from genes in flu virus strains in nonhumans for example, via a reasortment of genetic segments between viruses in humans and nonhumans rather than by mutations in annual human flu strains.

83. Which one of the following is the correct sequence in the order of decreasing length of the three structural parts given below of small intestine in the human body?

(a) Jejunum-Duodenum-Ileum
(b) Ileum-Duodenum-Jejunum
(c) Jejunum-Ileum-Duodenum
(d) Ileum-Jejunum-Duodenum

Ans. (d) : The correct sequence in the order of decreasing length of the three structural parts are Ileum – Jejunum – Duodenum of small intestine.

SCIENCE AND TECHNOLOGY

84. Which one of the following pairs is not correctly matched?

(a) Cosmic Background Explorer (COBE) : Satellite Programme
(b) Falcon : Under-sea cable system
(c) Discovery : Space Shuttle
(d) Atlantis : Space station

Ans. (d) : Atlantis was a Space Shuttle orbiter of National Aeronautics and Space Administration (NASA).

85. NASA's Deep Impact space mission was employed to take detailed pictures of which comet nucleus?

(a) Halley's Comet
(b) Hale-Bopp
(c) Hyakutake
(d) Tempel 1

Ans. (d) : NASA's Deep Impact space mission was employed to take detailed pictures of Tempel 1 comet nucleus. It is a periodic Jupiter-family comet that was discovered in 1867 by Wilhelm Tempel. As per the research, it completes an orbit of the sun every 5.5 years of time.

86. Where was the first conference of the Pugwash Conferences on Science and World Affairs held in the year 1957?

(a) Minnowbrook (USA)
(b) Rhode Island (USA)
(c) Nova Scotia (Canada)
(d) Nagasaki (Japan)

Ans. (c) : The Pugwash Conferences on Science and World Affairs is an international organization that brings together scholars and public figures to work toward reducing the danger of armed conflict and to seek solutions to global security threats. It was founded in 1957 by Joseph Rotblat and Bertrand Russell in Pugwash Nova Scotia Canada.

ENVIRONMENT AND ECOLOGY

87. How is Steve Fossett known as?

(a) As a crocodile hunter.
(b) For completing the longest nonstop flight around the globe.
(c) For swimming across Atlantic Ocean.
(d) For climbing to Mt. Everest without any co-climber.

Ans. (b) : Steve Fossett is known for completing the longest nonstop flight around the globe

CURRENT AFFAIRS

88. Which one of the following organisations won the CSIR Award for Science and Technology (S&T) innovations for Rural Development, 2006?

(a) CLRI (b) IARI
(c) NDDB (d) NDRI

Ans. (a) : Central Leather Research Institute (CLRI), received the first ever CSIR Award for S&T Innovations for Rural Development in 2006.

89. What is the Galileo Project which has been in news recently?

(a) An inter country programme of missile shield developed by the United States of America.
(b) A project developed by India with assistance from Canada.
(c) An environmental protection project being developed by Japan.
(d) A multi-satellite navigation project being developed by the European Union.

Ans. (d) : Galileo is the global navigation satellite system (GNSS) that is currently being created by the European Union (EU) and the European Space Agency (ESA).

90. Consider the following statements.

1. The series of the International Paper Sizes is based on A0 size whose area is 0.5 m^2 (approximately).
2. The area of A4 size paper is $1/8^{th}$ of that of the A0 size paper.

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (d) : The International Paper Quantity Series is based on the magnitude of A0, which has an area of 1.0 square meters (approximately). The area of A4 magnitude paper is $1/16^{th}$ of the area of A0 magnitude paper, not $1/8^{th}$. Hence both the given statements are wrong.